HANDS-ON · TAKE-HOME TELESCOPE

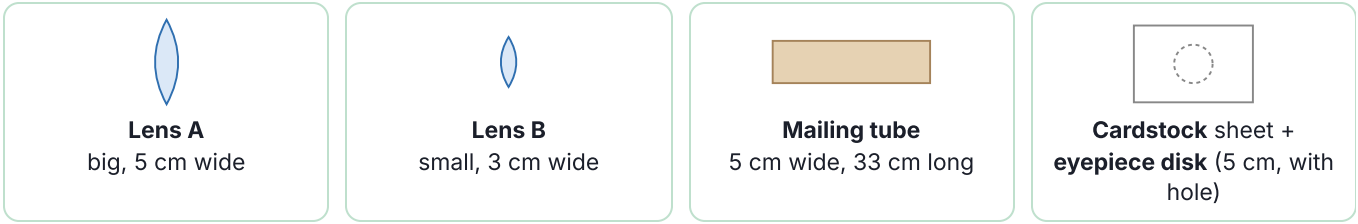
Build Your Telescope

Name _____

You've learned how lenses bend light, measured your two lenses, and built a telescope on a meter stick. Now you'll put the same two lenses in a tube so you can take your telescope home. Look for the green 🔗 **boxes**: each one connects a build step to what you learned in the lab.

Safety: never point your telescope at the Sun, not even for a second. It can permanently damage your eye. Put the warning sticker on your tube (Step 9).

0 Check your kit and your numbers



Lens A
big, 5 cm wide

Lens B
small, 3 cm wide

Mailing tube
5 cm wide, 33 cm long

Cardstock sheet + **eyepiece disk** (5 cm, with hole)

Also on your table: masking tape, a meter stick, a pencil, and a warning sticker.

FROM YOUR LAB WORKSHEET	YOUR NUMBER	WHY IT MATTERS
Lens A focal length f_o (Step 6)	cm	Lens A is the **objective** (front)
Lens B focal length f_e (Step 6)	cm	Lens B is the **eyepiece** (at your eye)
Spacing $f_o + f_e$ (Step 8)	cm	How far apart the lenses must be
Magnification $f_o \div f_e$ (Step 8)	×	How much bigger things will look

1 Get the big tube ready

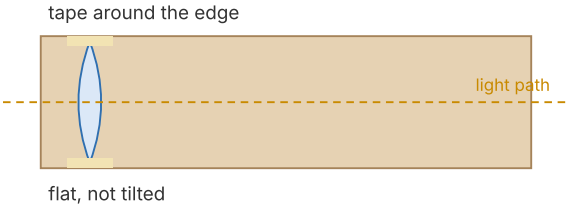
Pull both plastic caps off the mailing tube. Keep one: it's your lens cap. Look through the empty tube. With a pencil, write **FRONT** on one end.



> 🔗 **Lab Step 1.** Light travels in straight lines, so the tube doesn't bend or focus anything. It just holds the lenses lined up and blocks light from the side.

2 Mount Lens A (the objective) in the front

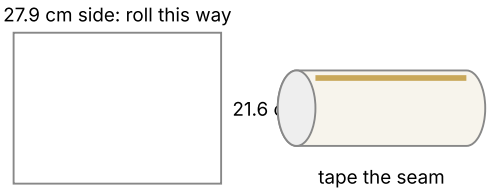
1. Wrap **one** layer of masking tape around the **edge** of Lens A, like a tire on a wheel. Don't tape over the glass.
2. Push the lens **straight** into the FRONT end, about 1 cm deep. It should fit snugly. Add a second layer of tape if it's loose.
3. Look at it from the side: the lens must be **flat across** the tube, not tilted.
4. Press a strip of tape around the inside rim in front of the lens to hold it in.



> 🔗 **Lab Steps 4, 6 and 9.** Lens A makes a small upside-down picture of faraway things about **30 cm** behind it (its focal length). It's your **light bucket**: 5 cm wide, so it collects about $(50 \div 7)^2 \approx$ **51 times** as much light as your eye.

3 Roll the sliding paper tube

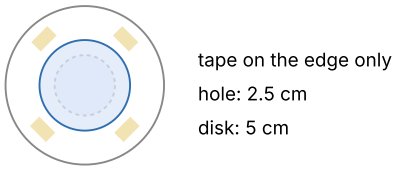
1. Hold the cardstock with the **short side (21.6 cm) up and down**. Roll it into a tube that is 21.6 cm long.
2. Slide the roll into the **back** of the big tube and let it spring open against the inside.
3. Pull it partway out and tape the seam along its whole length, keeping it that size.
4. Test: it should slide in and out with a gentle push, and not fall out on its own.



> 🔗 **Lab Step 8c and Step 11.** Sliding this tube changes the distance between the lenses. That's how you **focus**, just like sliding the eyepiece along the meter stick.

4 Make the eyepiece

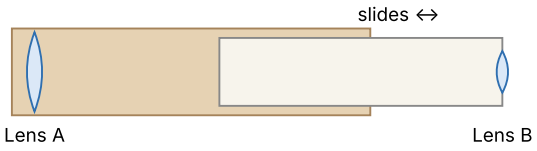
1. Center **Lens B** over the hole in the eyepiece disk.
2. Tape it down with 3 or 4 short pieces of tape over the **edge** of the lens only. Keep the middle of the glass clear.



> 🔗 **Lab Step 7.** Lens B has the **short** focal length (about 5 cm), so it's the strong magnifier: about 1 + 25 ÷ 5 = 6×. In your telescope it magnifies the picture Lens A makes.

5 Attach the eyepiece and assemble

1. Tape the eyepiece disk onto one end of the paper tube, lens centered.
2. Slide the other end of the paper tube into the **back** of the big tube, eyepiece end out.



> 🔗 **Lab Step 8.** Long-f lens in front, short-f lens at your eye. Swapped, your telescope would make things look **smaller**.

6 Predict where it will focus

The big tube holds Lens A about 32 cm from its back end. Your lenses need to be about $\mathbf{f_o + f_e}$ apart.

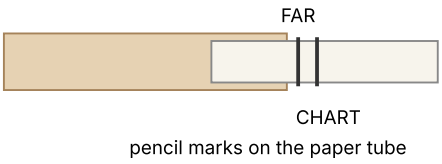
So the paper tube should stick out about (_____ + _____) − 32 = _____ cm past the end of the big tube.

Slide it to that spot and check with your meter stick: distance from Lens A to Lens B = _____ cm.

> 🔗 **Lab Step 8b.** The same plan you made for the meter-stick telescope: the eyepiece sits one eyepiece focal length behind the picture Lens A makes.

7 First light: focus it

1. Hold the telescope with both hands. Rest your elbows on a table or brace against a wall.
2. Put your eye **right behind** Lens B (about 1 cm away). Aim at the **far view out the window**. Tip: look over the tube at your target first, then drop your eye to the eyepiece.
3. Slide the paper tube **slowly** in and out until the view is sharp. Draw a pencil line on the paper tube where it meets the big tube and label it **FAR**.
4. Now aim at the **test chart** across the room and refocus. Draw a second line and label it **CHART**.



pencil marks on the paper tube

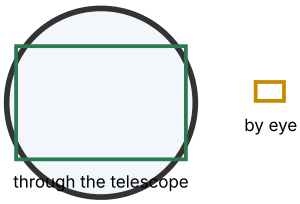
TARGET	PAPER TUBE STICKS OUT (CM)	THE VIEW IS
Far away (out the window)		☐ upside down ☐ right-side up
Test chart (across the room)		Smallest line I can read: _____ (by eye: _____)

Which needed the paper tube farther **out**? ☐ far view ☐ chart

> 🔗 **Lab Steps 5 and 11.** Closer objects make their picture **farther behind** Lens A, so the eyepiece has to move back to stay focused. And the view is upside down because Lens A's picture is upside down, just like on your card in Step 5.

8 Measure your magnification (two-eye trick)

Look at the test chart's **big square** through the telescope with one eye, and at the chart with your **other eye open** at the same time. The two views overlap. How many of the small (by-eye) squares fit across the big magnified one?
My telescope magnifies about _____ ×. My prediction (f_o ÷ f_e): _____ ×



> 🔗 **Lab Step 8.** Magnification = f_o ÷ f_e. With the kit lenses, 30 ÷ 5 = 6×.

9 Finish it

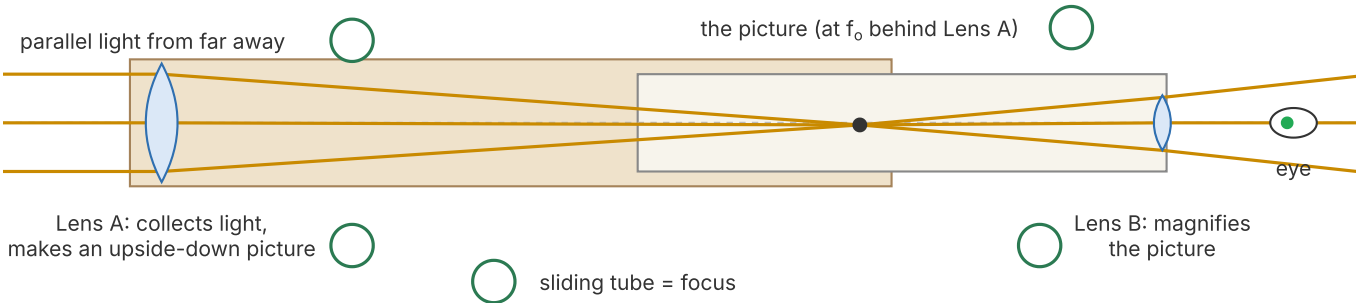
☐ Put the **"Never look at the Sun"** sticker on the big tube. ☐ Write your name on it. ☐ Put the cap on the front to protect Lens A.

Troubleshooting

PROBLEM	TRY THIS
I see nothing, or only a bright blur	Put your eye closer to Lens B and look through its **center**. Aim at something big and bright first.
It's blurry no matter where I slide it	Slide more slowly; the sharp spot is narrow. Check that Lens A is flat (not tilted) and both lenses are clean.
I run out of tube before it gets sharp	Lenses too close: pull the paper tube farther out, or push Lens A deeper into the big tube. Lenses too far apart: push the paper tube in, or trim the paper tube.
The view jiggles	Brace your elbows or rest the tube on a window ledge. Every wiggle is magnified 6×.
Colored edges on bright things	That's normal: glass bends colors by different amounts (Lab Step 10).

How your telescope works: put it together

Number each part of the diagram with the lab step that explains it.



Hint: choose from Lab Steps 1 (straight lines and parallel light), 4/6 (focal point), 5 (pictures), 7 (magnifier), 8 and 11 (telescope and focus).

Take it home: things to look at

- **The Moon.** Look along the line between the bright and dark parts: craters and mountains cast shadows there. Galileo saw these in 1609. Sketch what you see.
- **Jupiter's moons.** When Jupiter is in the evening sky (a free sky app like Stellarium will show you where), look for up to four tiny dots in a line beside it. Check again the next night: they move!
- **The Pleiades** (a star cluster high in the east on winter evenings). Count the stars by eye, then through your telescope.
- **Daytime:** a faraway sign, a bird in a tree, the bark on a distant tree. Remember: upside down, and **never** toward the Sun.

Home challenge. Focus on something far away, then on something across your room, and measure how far out the paper tube sticks each time. Use the lens equation from the page ($1 \div f = 1 \div d_o + 1 \div d_i$) to explain the difference.

Exit ticket. In your own words: what does Lens A (the objective) do? What does Lens B (the eyepiece) do? Why do astronomers build telescopes with **wide** lenses and mirrors?

One thing that surprised me today: